

Chapter 11 – The Opioids

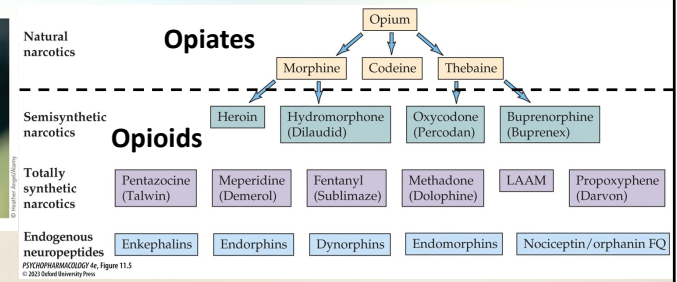
The Opioids

Chapter 11

Opiate narcotics are derived from a poppy plant

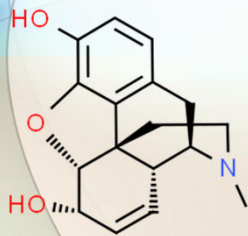


Papaver somniferum



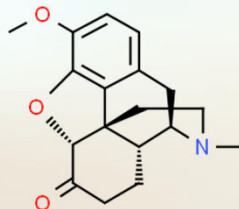
Opioid

Morphine



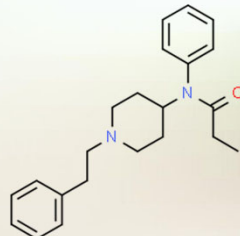
Semisynthetic narcotic

Oxycodone



Synthetic narcotic

Fentanyl



Endogenous opioids are derived from pro-peptides

POMC

- β -endorphin (β -END)

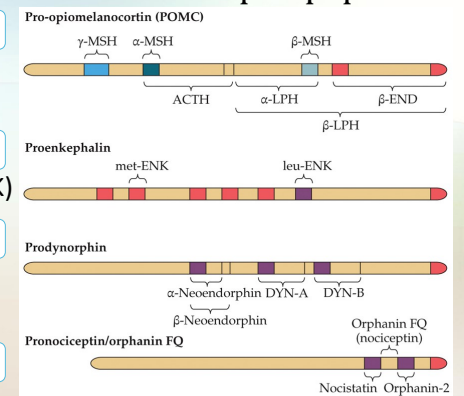
proenkephalin

- met- & leu-enkephalin (ENK)

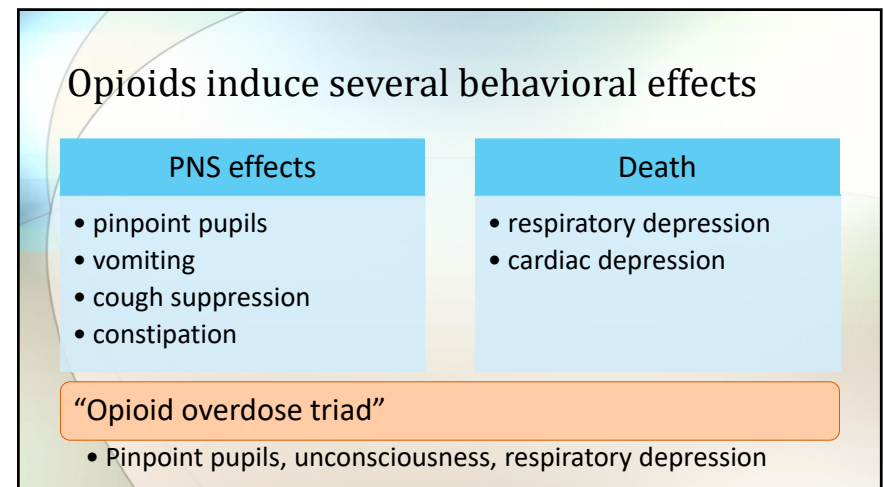
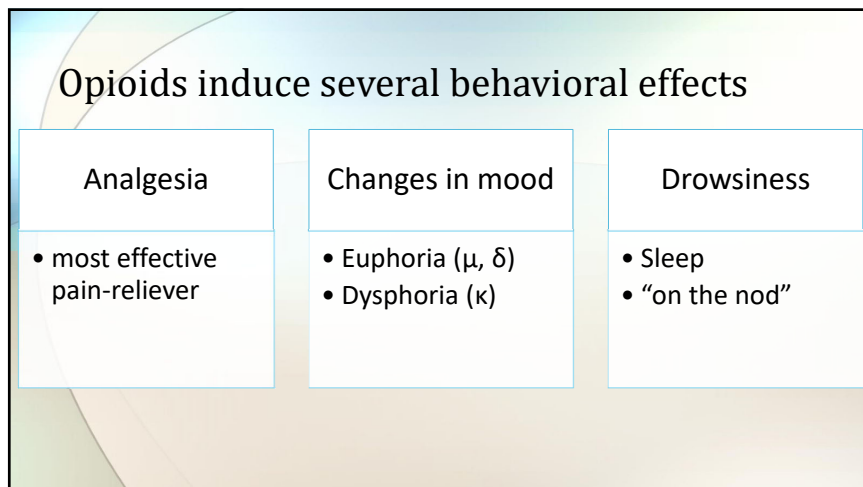
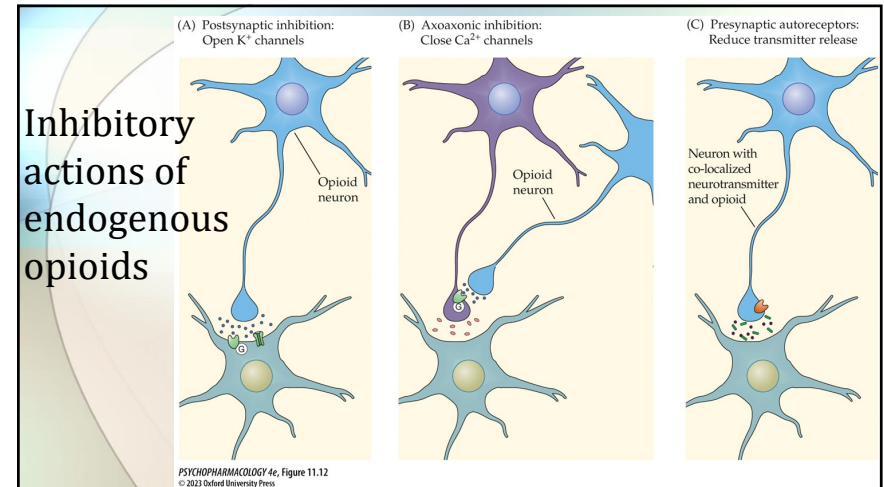
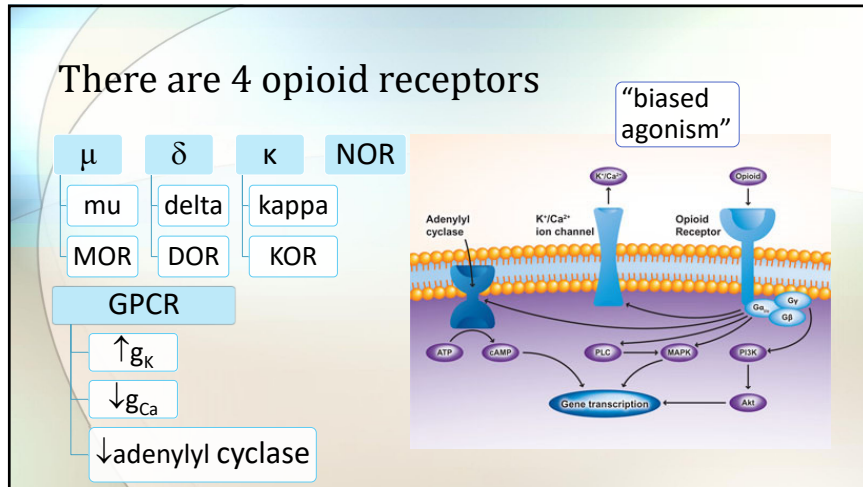
prodynorphin

- α - and β -neoendorphin
- dynorphin (DYN) A and B

pronociceptin/orphanin FQ



Chapter 11 – The Opioids



Chapter 11 – The Opioids

Acute MOR agonists are analgesic and reinforcing

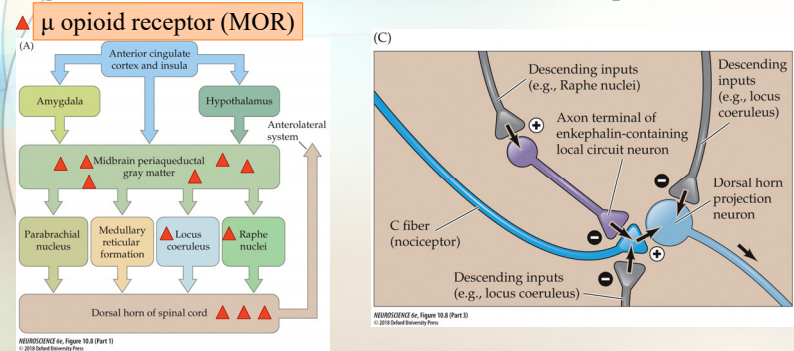
Analgesia

- Spinal
- Supraspinal

Reward

- Increased DA release in N. accumbens

Analgesic properties of opioids are due to agonism of MOR located in brain and spinal cord



MOR agonists activate mesolimbic DA neurons and have reward properties

μ Opioid Agonists

Intracranial self-stimulation (ICSS)

Self-administration

Microinjection studies

MOR agonists activate mesolimbic DA neurons and have reward properties

μ Opioid Agonists

Intracranial self-stimulation (ICSS)

- Opioids decrease the threshold current required for ICSS

Chapter 11 – The Opioids

MOR agonists activate mesolimbic DA neurons and have reward properties

μ Opioid Agonists



Self-administration

- IV use gradually increases over time; similar to the pattern seen in humans

MOR agonists activate mesolimbic DA neurons and have reward properties

μ Opioid Agonists



Microinjection studies

Intra-VTA microinjection

- $\uparrow$ firing rate and DA release from VTA
- induces CPP
- $\downarrow$ threshold for ICSS

MOR agonists activate mesolimbic DA neurons and have reward properties

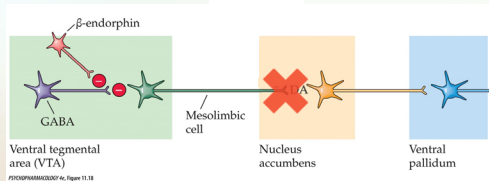
- Opioids disinhibit VTA neurons, increasing their firing rate, causing more DA release in the N. Acc.

GPCR

$\uparrow g_K$

$\downarrow g_{Ca}$

$\downarrow$ adenylyl cyclase



- 6-OHDA lesions reduce self-administration, showing the rewarding properties of opioids also involve other, non-DA mechanisms

MOR agonists activate mesolimbic DA neurons and have reward properties

μ Opioid Agonists



Intracranial self-stimulation (ICSS)

- Opioids decrease the threshold current required for ICSS



Self-administration

- IV use gradually increases over time; similar to the pattern seen in humans



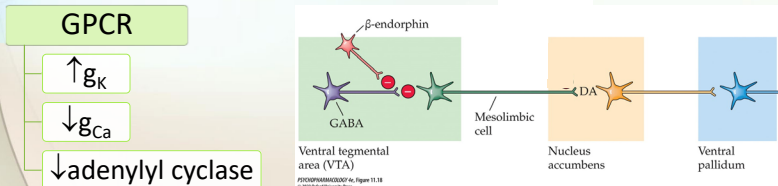
Microinjection studies

- $\uparrow$ firing rate & DA release from VTA; induces CPP; $\downarrow$ threshold for ICSS

Chapter 11 – The Opioids

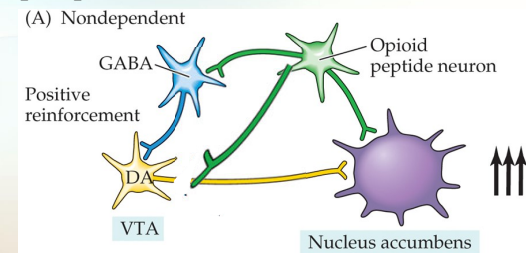
MOR agonists activate mesolimbic DA neurons and have reward properties

- Opioids disinhibit VTA neurons, increasing their firing rate, causing more DA release in the N. Acc.



MOR agonists activate mesolimbic DA neurons and have reward properties

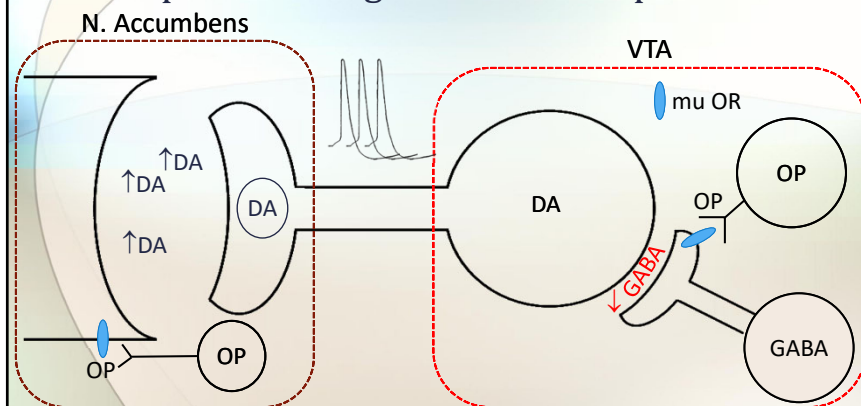
- Opioids disinhibit VTA neurons, increasing their firing rate, causing more DA release in the N. Acc.



After G. F. Koob and M. Le Moal, 2008. *Ann Rev Psychol* 59: 29–53. Republished with permission of Annual Reviews, Inc.

PSYCHOPHARMACOLOGY 4e, Figure 9.18 (Part 1)
© 2013 Oxford University Press

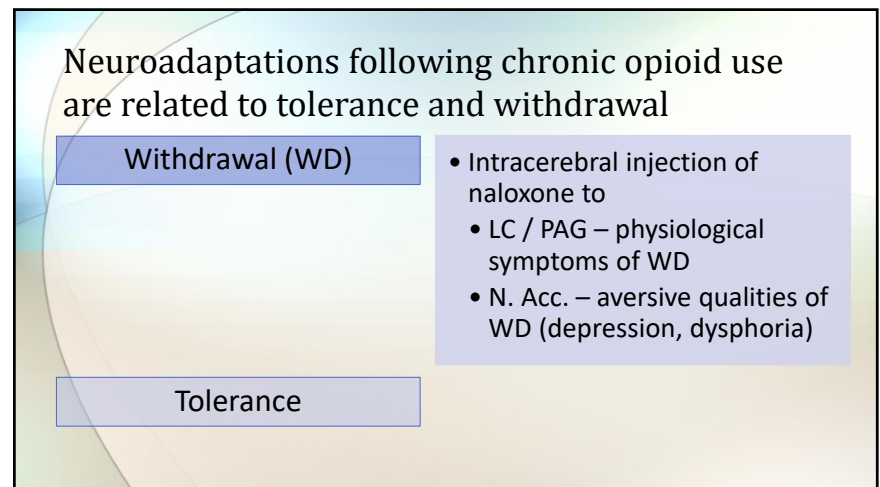
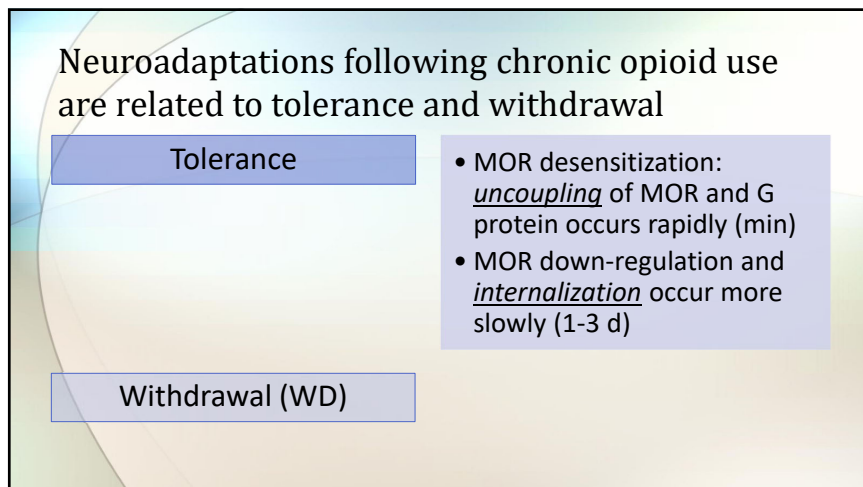
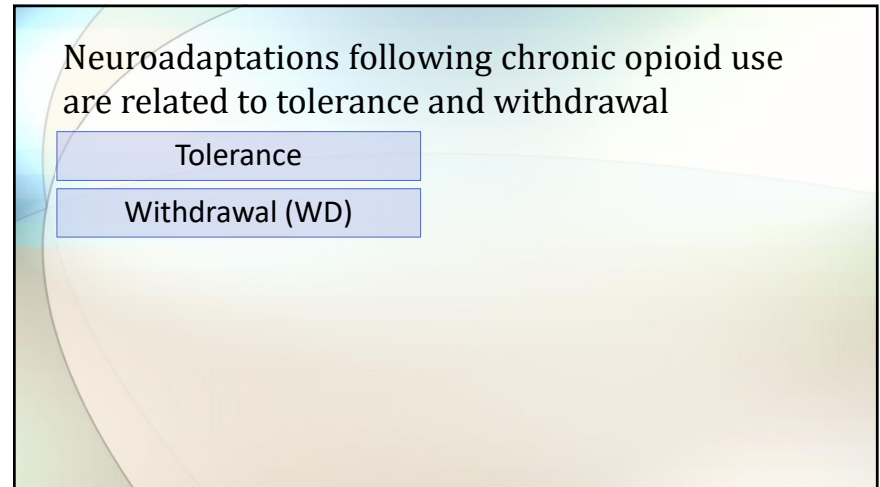
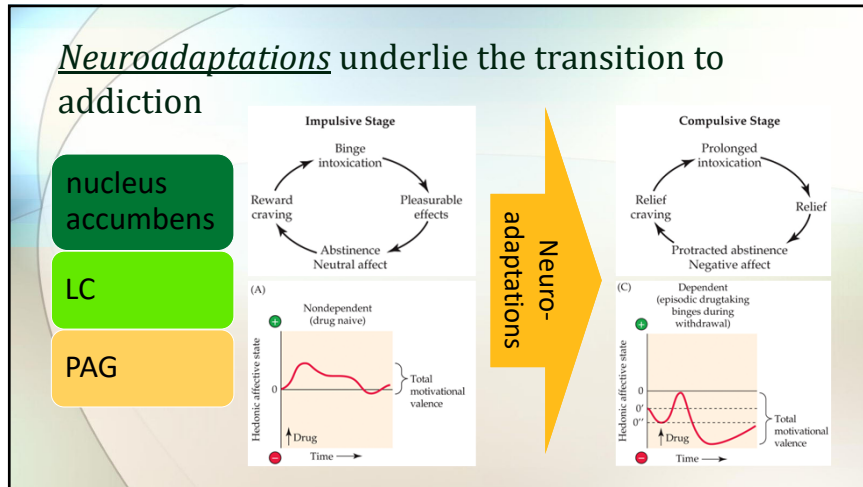
Acute pharmacological effects of opioids



Long-term opioid use results in tolerance & cross-tolerance; incentive sensitization; physical dependence

Acute action	Withdrawal sign
Analgesia	Pain and irritability
Respiratory depression	Panting and yawning
Euphoria	Dysphoria and depression
Tranquilization	Fearfulness and hostility
Decreased blood pressure	Increased blood pressure
Constipation	Diarrhea
Pupil constriction	Pupil dilation
Hypothermia	Hyperthermia
Drying of secretions	Tearing, runny nose
Reduced sex drive	Spontaneous ejaculation
Flushed and warm skin	Chilliness and "gooseflesh"

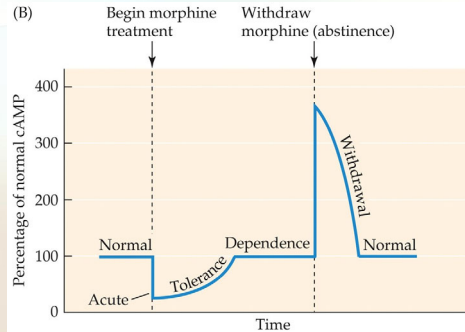
Chapter 11 – The Opioids



Chapter 11 – The Opioids

Abrupt removal of opioid in tolerant /dependent animals leads to a withdrawal syndrome

Himmelsbach's model



Neuroadaptations following chronic opioid use are related to tolerance and withdrawal

Acute

- activate MOR in LC
- Hyperpolarization; less AP firing

Chronic

- LC neurons gradually increase AP firing
- cAMP / PKA activity gradually increases

Induce WD

- administer naloxone to dependent animal
- LC neurons are over-excited
- cAMP / PKA over-activated

Summary: Neurobiology of Addiction

